

SOFTWARE REQUIREMENTS SPECIFICATION

SIM — Security Incident Mapping

A Web Platform for Campus Security Incident Reporting

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the Security Incident Mapping (SIM) system, a web-based platform designed to allow students, staff, and landlords within the campus community to report security incidents quickly and securely. This SRS describes the functional and non-functional requirements of the system and is intended for the course instructor, evaluators, and future developers of the platform.

1.2 Scope

SIM is a lightweight web application that enables members of the campus community to report incidents such as theft, assault, harassment, stabbing, and other security concerns. The system captures the incident type, location, severity, and description, with an option to report anonymously. Reports are logged and made visible in a live incident feed, which can be filtered by incident type. The system is aimed at reducing the underreporting of campus crime by removing common barriers to reporting, such as fear of retaliation and the absence of a fast, standardized reporting channel.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SIM	Security Incident Mapping — the name of the system
SRS	Software Requirements Specification
Actor	A user or role that interacts with the system
Use Case	A specific action or interaction an actor can perform with the system
Anonymous Report	A report submitted without identifying the reporter

1.4 References

- SEN202 Weekend 1 Dev Assignment brief, Dr. O. O. Ajayi (01/07/2026)
- SIM Google Form Questionnaire (student/staff/landlord needs assessment)
- SIM Ishikawa (Fishbone) Diagram — root cause analysis of campus insecurity
- SIM Mini Web Interface (index.html) — proof-of-concept implementation of the system
- SIM Use Case Diagram — actor and system interaction model

1.5 Overview

Section 2 of this document describes the overall context and background of the system. Section 3 details the functional requirements. Section 4 details the non-functional requirements. Section 5 summarizes the system models (use case diagram and Ishikawa analysis) that informed this specification.

2. Overall Description

2.1 Product Perspective

SIM is a standalone, self-contained web interface. It does not depend on any existing campus security infrastructure and is designed as a proof-of-concept mini web application, in line with the project's constraint to avoid unnecessary complexity given the submission timeline. It is intended to be extended in future iterations with real-time mapping, notification systems, and integration with campus security response teams.

2.2 Product Functions

At a high level, SIM allows the following:

- Reporting a security incident, including type, location, severity, and description
- Choosing to report anonymously or with a role identified (student, staff, or landlord)
- Viewing a live log of all reported incidents
- Filtering the incident log by incident type
- Registering a landlord's property location for future mapping and faster response

2.3 User Classes and Characteristics

User Class	Description
Student	Lives on or off campus; primary reporter of incidents witnessed or experienced
Staff	Works within the institution; reports incidents and relies on institutional response procedures
Landlord	Owens off-campus student housing; reports incidents at their property and registers property location
Security Admin	Views and manages submitted reports; monitors incident patterns

2.4 Operating Environment

SIM runs as a web application accessible through any modern mobile or desktop browser. No installation is required. It does not require a persistent server backend in its current mini-application form; incident data is retained locally within the user's browser session for demonstration purposes.

2.5 Design and Implementation Constraints

- The system must remain simple in design, given the limited submission timeframe (constraint stated in the project brief).
- The system must support anonymous reporting to address the sensitivity of the subject matter.
- The system must be usable primarily from a mobile device, as this is the primary access point for most users in the target community.

3. Functional Requirements

The following functional requirements are derived directly from the SIM use case diagram and reflect the actual behaviour implemented in the mini web interface.

FR1 — Select Role

The system shall allow a user to identify their role (Student, Staff, or Landlord) before submitting a report.

FR2 — Submit Incident Report

The system shall allow any user to submit an incident report consisting of incident type, location, severity, and a free-text description.

FR3 — Choose Incident Type

The system shall provide a predefined list of incident types (Theft, Assault, Harassment, Stabbing, Murder, Suspicious Activity, Other) for the user to select from when submitting a report.

FR4 — Report Anonymously

The system shall allow the user to optionally submit a report without disclosing their role or identity.

FR5 — View Incident Log

The system shall display a chronological log of all submitted incident reports, visible to all users.

FR6 — Filter Reports by Type

The system shall allow a user to filter the incident log to display only reports of a selected incident type.

FR7 — Register Property Location

The system shall allow a landlord to register the location of their property in relation to the campus for future mapping purposes.

FR8 — Manage and Resolve Reports

The system shall allow a Security Admin to view all submitted reports and mark incidents for follow-up or resolution.

ID	Requirement	Priority
FR1	Select Role	High
FR2	Submit Incident Report	High
FR3	Choose Incident Type	High
FR4	Report Anonymously	High
FR5	View Incident Log	High
FR6	Filter Reports by Type	Medium
FR7	Register Property Location	Medium
FR8	Manage and Resolve Reports	Medium

4. Non-Functional Requirements

NFR1 — Usability

The system shall be simple enough for a first-time user to submit a report in under 2 minutes without prior training, given that most users access it from a mobile device.

NFR2 — Privacy and Anonymity

The system shall not require personal identifying information (such as name, email, or phone number) for a report to be submitted, in order to encourage honest reporting of sensitive incidents.

NFR3 — Availability

The system shall be accessible at any time via a web browser, without requiring account creation or login.

NFR4 — Performance

The system shall update the incident log and filtered views within 1 second of a report being submitted or a filter being applied.

NFR5 — Portability

The system shall function correctly on both mobile and desktop web browsers without requiring a separate native application.

NFR6 — Maintainability

The system shall be implemented using a single, self-contained HTML file to minimize complexity and ease future maintenance, in line with the project's development constraints.

ID	Requirement	Category
NFR1	Usability — under 2 minutes to submit a report	Usability
NFR2	No personal data required to submit a report	Privacy/Security
NFR3	Accessible anytime, no login required	Availability
NFR4	Log updates within 1 second	Performance
NFR5	Works on mobile and desktop browsers	Portability
NFR6	Single-file implementation for maintainability	Maintainability

5. System Models

5.1 Use Case Summary

The system involves four actors — Student, Staff, Landlord, and Security Admin — interacting with core use cases including Submit Incident Report, View Incident Log, and Manage/Resolve Reports. The 'Submit Incident Report' use case includes the mandatory sub-steps 'Select Role' and 'Choose Incident Type', and optionally extends to 'Report Anonymously'. The full use case diagram is provided as a separate artifact (SIM_UseCase_Diagram.drawio) in the project repository.

5.2 Root Cause Analysis Summary

The Ishikawa (fishbone) diagram identified six categories of root causes contributing to underreported campus insecurity: People, Process, Environment, Technology, Policy/Management, and Materials/Resources. Several of these causes directly informed the functional requirements above — for example, 'fear of retaliation' (People) informed the anonymous reporting requirement (FR4/NFR2), and 'no standardized reporting procedure' (Process) informed the structured report submission flow (FR2, FR3). The full diagram is provided as a separate artifact (SIM_Ishikawa_Diagram.drawio) in the project repository.

5.3 Traceability

Source	Informs
Google Form — fear of reporting barrier (staff/students)	FR4 — Report Anonymously
Ishikawa — no standardized reporting procedure	FR2, FR3 — Structured report submission
Ishikawa — no mapping/geolocation of hotspots	FR7 — Register Property Location
Google Form — landlord willingness to cooperate	FR7 — Register Property Location
Ishikawa — no follow-up system after report is made	FR8 — Manage and Resolve Reports